Bellabeat Smart Device Analysis

Intro

This case study explores how consumers use their smart fitness devices, using publicly available Fitbit data from 30 users over the course of 2 months. I cleaned and consolidated several CSV files in Google Sheets, then built an interactive Tableau dashboard to examine four angles of user behavior: when users move throughout the day, how consistent the patterns are across the week, how activity levels distribute across the population, and whether tracking frequency correlates with activity. The analysis surfaced 3 findings: users follow distinct weekday and weekend rhythms, over a third of all tracked days are sedentary, and app engagement does not predict physical activity. I translated these findings into actionable recommendations for Bellabeat's executive team. The project demonstrates end-to-end data analysis: from data cleaning and integrity checks, through exploratory visualization, to business-facing recommendations grounded in evidence.

Ask

Business task: Analyze Fitbit usage data to determine how consumers use non-Bellabeat smart devices, then reveal growth opportunities for Bellabeat's marketing strategy.

Key stakeholders:

- Urška Sršen, cofounder and Chief Creative Officer
- Sando Mur, cofounder and executive team member

- Bellabeat marketing analytics team

Guiding questions:

1. What are some trends in smart device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat's marketing strategy?

Prepare

The primary data source for this analysis is the FitBit Fitness Tracker Data dataset, made publicly available on Kaggle under a CCO: Public Domain license and originally contributed by Mobius (hosted at kaggle.com/datasets/arashnic/fitbit). The dataset contains personal fitness tracker data from thirty eligible Fitbit users who consented to the submission of their personal tracker data, collected via Amazon Mechanical Turk.

The data is organized into two folders corresponding to two overlapping time windows:

- **March 12 – April 11, 2016** (first collection window)
- **April 12 – May 12, 2016** (second collection window)

Across both windows, the dataset contains 18 CSV files covering a range of tracking dimensions: daily activity summaries, hourly steps, hourly calories, hourly intensities, minute-level sleep, heart rate, METs, and weight logs. The files come in both "narrow" and "wide" formats depending on the metric. After reviewing the available files, I selected three primary sources for the analysis: dailyActivity_merged, hourlyIntensities_merged and hourlyCalories_merged. Files that are missing from one of the collection windows (such as dailyCalories_merged and dailyIntensities_merged) were either excluded or replaced by equivalent hourly aggregations.

Does the data ROCCC?

- **Reliable:** There are only 30 respondents in this dataset. This is a really small sample size and may come with biases. Additionally, the data lacks entries such as gender and age. This is not helpful for BellaBeat as the company focuses on women's health specifically.
- **Original:** Partially. The data was not collected by Bellabeat or by this analysis; it is third-party data collected from Fitbit devices and aggregated by Mobius. This introduces some uncertainty about collection methodology and device consistency, though the data appears internally consistent across files.
- **Comprehensive:** Partially. The first batch of data is missing multiple CSV files. Some data can be extrapolated such as calculating the missing *dailyCalories_merged.csv* file by using the *hourlyCalories_merged.csv* file.
- **Current:** The data is over a decade old, from 2016. Smart fitness devices have evolved greatly during the last decade, making the data very outdated.
- **Cited:** The data was collected via Amazon Mechanical Turk and is publicly documented and licensed.

Process

I chose Google Sheets for cleaning and consolidating, and Tableau for the analysis and visualization. Google Sheets was the right choice for cleaning because the raw files are small enough to open directly, the data required mostly manual formatting fixes, and Sheets allowed me to quickly inspect and correct date/time issues without writing code. Tableau was chosen for the analysis phase because it handles multi-dimensional grouping (day × hour × category) far more gracefully than spreadsheets, and it made the interactive dashboard possible.

Cleaning Steps

1. File consolidation. The March–April and April–May folders contain overlapping but incomplete sets of files. I combined both windows into unified sheets so that

each metric (daily activity, hourly intensity, hourly calories) existed as a single continuous table across the full two-month range. This step alone eliminated the need to join windows during analysis.

2. Duplicate removal. I checked each combined sheet for duplicate rows and removed them. Duplicates are common when merging overlapping time windows, and leaving them in would have double-counted activity in the pivot-level aggregations.

3. Blank and missing value inspection. I applied conditional formatting to highlight blank cells, then reviewed the flagged rows manually to determine whether the blanks represented genuine missing data or a formatting artifact. In nearly all cases the blanks were structural (e.g., hours with no activity recorded), not data errors, so they were preserved rather than deleted.

4. Date and time reformatting. The most substantial cleaning step was fixing date and time formats. In the raw files, dates and timestamps were stored as text strings, which prevents sorting, filtering, and time-based grouping. I reformatted:

- Plain date strings → proper date values (e.g., 3/12/2016)
- Combined date/time strings → proper date-time values
- Then split the date-time values into separate Date and Time columns, so the analysis could group by day without dragging the timestamp along, and separately group by hour within a day.

This step was essential — without it, the hourly heatmap and the weekday/weekend curve in the Share section would not have been possible in Tableau.

5. Column standardization. Where column names differed slightly between the two windows (spacing, capitalization, or suffix), I standardized them so the combined sheets had consistent headers. This made it possible to use a single pivot table and a single Tableau data source per metric.

6. Categorization logic. For the distribution analysis, I added a new column called ActivityLevel that classifies each tracked day into one of five standard tiers based on total steps:

Category	Step range
Sedentary	< 5,000
Low active	5,000 – 7,499
Somewhat Active	7,500 – 9,999
Active	10,000 – 12,499
Highly active	12,500+

I also created a numeric ActivityScore (Sedentary = 1 through Highly active = 5) to support the regression in Finding 4, and a DayType column (Weekday / Weekend) and DaySort column (Monday = 1 through Sunday = 7) to support the time-series and heatmap visualizations. These helper columns live in the cleaned data and are carried into Tableau as dimensions.

7. Exclusion decisions. As noted in the Prepare section, weightLogInfo_merged was excluded because only about one-third of users had entries. Sleep data was excluded from the main findings because it only exists for one of the two time windows.

Data Integrity Verification

To verify the cleaned data was ready for analysis, I checked the following:

- Row counts matched expectations from the raw source (allowing for documented duplicate removals).
- Date ranges were continuous across the two combined windows with no unexplained gaps.
- Participant count was verified using COUNTUNIQUE on the Id column to confirm all 30 users were present after consolidation.
- Spot-checked values on a sample of rows against the raw files to confirm no accidental transposition or formatting errors were introduced during consolidation.

- Pivot-level totals (steps, calories, intensity) were compared against hourly-level aggregations rolled up to daily to ensure the two views agreed.

Deliverables

The cleaned data was exported as a consolidated summary file and loaded into Tableau as the source for all four visualizations. The cleaning steps documented above are reproducible from the raw Kaggle files using the same sequence.

Analyze

After cleaning and consolidating the data, I set out to answer how consumers actually use smart fitness devices, and how this data could reveal growth opportunities for Bellabeat. I explored users' activity patterns (when they move), their day to day consistency, distribution, and their engagement (whether tracking frequency correlates with activity). Each finding is supported with a visualization and framed for marketing implications for Bellabeat.

Finding 1: Smart device users follow a distinct weekly activity rhythm

To understand when users are most active throughout the day, I plotted average total intensity by hour of day, split between weekdays and weekends. All 30 users are aggregated across both months and reveal a pattern. During the week, activity rises between 4:00 and 5:00 AM, climbs steadily through the morning, and peaks between 5:00 and 6:00 PM, with a notable lull at around 3:00 PM. Weekend activity, by contrast, starts at around 6:00-7:00 AM and hits its peak earlier, around 11:00 AM - 1:00 PM. The levels remain high with no dip in the afternoon, with both curves declining together after 8:00 PM, dropping to baseline at 11:00 PM. Bellabeat could strategically notify users throughout the day, nudging them to

move when they typically become active (depending on the day of the week), or a reminder to move around during the weekday dip at 3:00 PM.

Finding 2: Sleep and wake patterns are highly consistent across the week

To examine how activity is distributed across both the hours of the day and the days of the week simultaneously, I built a heatmap with the hour of the day on the horizontal axis and the day of the week on the vertical axis. Each cell shows the average total intensity for that hour on that day, with darker shading indicating higher activity. The data shows 3 patterns of activity. The overnight block, between 12:00 AM and 4:00 AM, shows a drop to baseline during these hours regardless of whether it's a Monday or a Saturday. One would assume that most users sleep in on the weekends, but the trend shows a consistent, stable sleep schedule. Between 5:00 AM and 8:00 AM is where there is a huge contrast between weekdays and weekends. During the week, there is a sharp ramp up at around 5:00 AM, while weekends remain less active until 7:00 AM, reinforcing the results from the first visualization. Sunday is the least active day overall, with Saturday being one of the most active days between 9:00 AM and 12:00 PM. This contradicts a common assumption that weekends are uniformly more active than weekdays. The data shows that Saturday and Sunday are very different.

Finding 3: The majority of tracked days fall into the Sedentary category

To understand how activity levels distribute across the population, I categorized every tracked day into one of five standard activity tiers (Sedentary, Low active, Somewhat active, Active, Highly active) and calculated the percentage of all tracked days falling into each tier. Sedentary days make up 35.79% of the distribution, more than double any of the other categories. This is followed by Low active (18.18%) and Active (16.54%), as well as Somewhat active (15.25%) and

Highly active (14.24%). A bimodal curve is revealed, spiking at Sedentary and Active, with Somewhat active dipping in the middle. Users who reach 10,000 steps do not tend to stop at 8,000, those that commit to the target hit it or fall well short of it. With such a large portion of the population fitting into the Sedentary category, Bellabeat has an opportunity to target reducing sedentary time. With the bimodal nature of the data, there are 2 distinct user archetypes, the "all in" user that hits 10K+ steps consistently, and the "checked out" user who falls short of 5,000. Both groups should have a distinct campaign.

Finding 4: Tracking frequency does not predict activity level

To test a common assumption, that the most engaged users are the healthiest, I examined the relationship between how often each user logged entries and how active they were on average. I assigned each tracked day a numeric activity score between 1 (Sedentary) to 5 (Highly active), then averaged that score per user and plotted it against the user's total entry count. There is a weak positive curve that is statistically negligible. A linear regression produces an R^2 of 0.077, meaning that tracking frequency only accounts for about 8% of the variance in average activity level, which means 92% of the variation comes from something else entirely. Interestingly, the highest frequency trackers clustered toward the low end of activity. This null result is useful, because the overturning of this assumption saves Bellabeat marketing funds from upselling the more frequent users on subscriptions, since the most frequent trackers tended to be the least active. This also suggests that engagement and activity is independent, and the most prolific trackers may actually be users that monitor their health rather than perform, either because they are trying to change their behavior or because they track passively through automatic syncing.

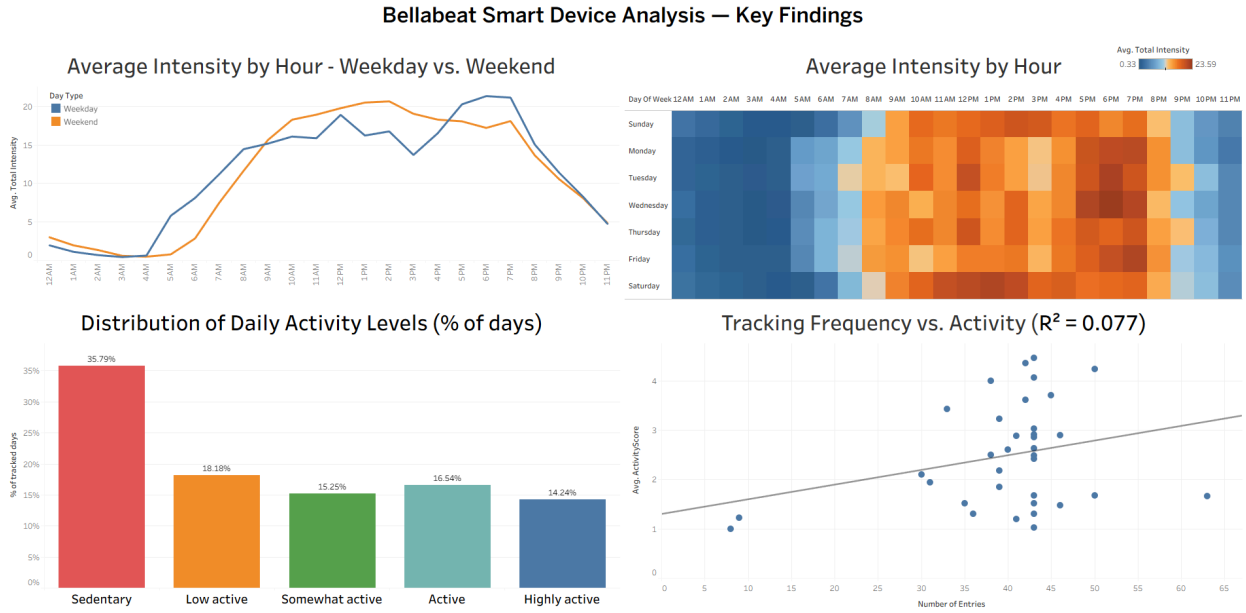
Summary of Analysis

Through these findings, a pattern emerges. The users are active, but they are active inconsistently. Both the weekday/weekend curve and the heatmap show that there are clear and predictable windows for activity, but they are narrow and easy

to miss. The activity distribution confirms this, with over a third of all tracked days falling into the Sedentary category, and the distribution is bimodal rather than gradual. The users will either commit to 10,000 steps or fall well short of that goal, with little middle ground. The findings from the tracking vs. activity scatter plot highlighted that the tracking frequency does not predict physical activity, meaning Bellabeat can't assume that the most loyal users are also the most active ones. A key opportunity is found with these findings: It is more advantageous to help users improve their worst days rather than push for peak performance, and the suggestion to send timed notifications was revealed.

Share

The goal of the Share phase is to give the Bellabeat executive team a clear, memorable picture of user behavior that supports specific marketing decisions. I designed four visualizations, each answering a distinct question and each building toward the recommendations in the Act section.



[Tableau Public Interactive Dashboard]

Figure 1 — Weekday vs. Weekend Average Intensity by Hour (line chart)
This line chart plots average total intensity against hour of day, with two lines representing weekdays and weekends. It was chosen over a bar chart because the question is about *shape over time*, and a line conveys continuity and rhythm more naturally than discrete bars. The chart's primary visual anchor is the divergence between the two lines in the morning hours.

Figure 2 — Average Intensity Heatmap by Day of Week and Hour (heatmap)
This heatmap places hours on the horizontal axis and days of the week on the vertical axis, shading each cell by average intensity. A diverging color scale (light to dark) is used so high-activity windows stand out without requiring axis comparison. This visualization compresses two dimensions (day and hour) into a single glanceable image. Its strength is pattern recognition: readers can identify the overnight sleep block, the weekday morning ramp, and the Sunday anomaly in seconds.

Figure 3 — Distribution of Daily Activity Levels (bar chart)
This bar chart shows the percentage of tracked days falling into each of the five activity categories, ordered from Sedentary to Highly active. The bars are colored on a red-to-blue gradient matching the natural severity of each category, so the visual language reinforces the finding without needing a legend. Data labels appear above each bar and the Y-axis is formatted as a percentage. The design draws the eye to the first bar: Sedentary at 35.79% is nearly twice the height of any other.

Figure 4 — Tracking Frequency vs. Average Activity Score (scatter plot)
This scatter plot places each of the 30 users as a single point, with tracking frequency on the X-axis and average activity score (1–5) on the Y-axis. A linear trend line is overlaid to show the weak positive relationship ($R^2 = 0.077$). A scatter was chosen because it is the only chart type that displays individual-level variation while also supporting a regression-based conclusion. With only 30 users, each dot needs to be visible — and the overall pattern (or lack thereof) must be legible at a glance.

Act

Recommendations for Bellabeat Marketing Strategy

The analysis surfaces a consistent theme: Bellabeat's users have clear, predictable activity rhythms, but they are not consistently active within those rhythms. The following three recommendations translate the findings into concrete marketing and product actions. Each is tied to a specific visualization and a specific finding, and each is designed to be implementable within Bellabeat's existing product ecosystem (Leaf, Time, Spring, and the Bellabeat app).

Recommendation 1: Move from a fixed notification schedule to a day-aware, rhythm-based engagement model.

Supported by: Finding 1 (Weekday vs. Weekend curve) and Finding 2 (Heatmap)

The Weekday vs. Weekend curve shows that weekday activity begins rising sharply between 4:00 and 5:00 AM and peaks between 5:00 and 6:00 PM, while weekend activity doesn't begin rising until 6:00–7:00 AM and peaks much earlier in the day. The heatmap confirms this pattern is stable across every week in the dataset.

Action: Bellabeat should schedule app notifications, motivational messages, and gentle movement prompts based on each user's *observed* activity rhythm, not on a uniform clock. A morning nudge at 7:00 AM will consistently reach weekday users as they begin their day, but the same nudge on a Saturday morning will arrive before most users are awake. A simple first step is a rule-based system that adjusts notification timing by day type: earlier on weekdays, later on weekends. A second-phase improvement would personalize timing per user based on their historical activity curves.

Why it matters: Timed nudges have substantially higher engagement rates than generic ones. Sending a "time to move" message at the moment a user is already

inclined to move is the single lowest-cost, highest-impact change Bellabeat can make.

Recommendation 2: Reframe the primary marketing message around sedentary reduction, not peak step goals.

Supported by: Finding 3 (Activity distribution)

The activity distribution shows that **35.79% of all tracked days fall into the Sedentary category**, which is nearly double any other tier. Only 14.24% reach Highly active status. The chart also reveals a bimodal pattern: users are either committing to 10,000+ step days or barely moving at all, with very little middle ground.

Action: Bellabeat's marketing should shift its primary message from "hit 10,000 steps" (which only ~30% of tracked days achieve) to **"break up your sitting time"** or **"every hour counts."** This is a lower barrier to entry, aligns with the actual behavioral data, and differentiates Bellabeat from Fitbit and Apple Watch, both of which lean heavily on aspirational step goals. Feature-level implementation could include hourly movement reminders during the user's typical sedentary windows, a "sedentary streak" counter, or small rewards for breaking up long periods of inactivity.

Why it matters: Marketing messages that meet users where they actually are convert far better than messages that ask them to be someone they aren't yet. Bellabeat has a real opportunity to own the "small, consistent movement" category in the women's wellness space, a category that is currently underserved by competitors.

Recommendation 3: Segment users on two axes, engagement and activity, rather than treating them as one spectrum.

Supported by: Finding 4 (Tracking vs. Activity scatter)

The scatter plot shows that tracking frequency explains only about 8% of the variance in average activity score ($R^2 = 0.077$). Users who log entries most frequently are not more active, in fact, the highest-frequency trackers cluster

toward the *lower* end of the activity scale. This overturns a common marketing assumption that the most engaged app users are also the healthiest customers.

Action: Bellabeat should build a **two-axis user segmentation model**: one axis for engagement (frequency of app use) and one for activity (average movement level). This produces four distinct user archetypes:

- **High engagement + high activity** — already aligned; upsell to premium membership for advanced analytics.
- **High engagement + low activity** — the most promising cohort for behavioral change interventions; they're *trying* but not yet moving.
- **Low engagement + high activity** — need a reason to open the app; gamification and social features work well here.
- **Low engagement + low activity** — need re-engagement campaigns before any other intervention.

Why it matters: A single-axis "engagement score" would misclassify users and waste marketing budget. The two-axis model lets Bellabeat personalize messaging by archetype and allocate campaign spend to the cohorts most likely to respond. The highest-value cohort — high engagement + low activity — is currently invisible under a one-axis model.

Limitations and Next Steps

This analysis has notable limitations. The 30-user sample is small and demographically opaque. No gender or age data was available, which is a particular weakness for a women's wellness company. The data is nearly a decade old, and smart device usage patterns have shifted substantially since 2016. The dataset also lacks Bellabeat-specific users, meaning all findings are inferences about behavior rather than observations of Bellabeat's actual customer base.

Future work should incorporate a larger, gender-disaggregated dataset and validate the recommendations through A/B testing on Bellabeat's own user base. The two-axis segmentation model in particular would benefit from being tested against real cohort-level data before being adopted at scale.

Portfolio Summary

This case study explores how consumers actually use smart fitness devices, using publicly available Fitbit data from 30 users over two months. Acting as a junior data analyst on Bellabeat's marketing team, I cleaned and consolidated 18 CSV files in Google Sheets, then built an interactive Tableau dashboard to examine four angles of user behavior: when users move throughout the day, how consistent those patterns are across the week, how activity levels distribute across the population, and whether tracking frequency correlates with activity. The analysis surfaced three actionable findings — users follow distinct weekday and weekend rhythms, over a third of all tracked days are sedentary, and app engagement does not predict physical activity — which I translated into three marketing recommendations for Bellabeat's executive team. The project demonstrates end-to-end data analysis: from data cleaning and integrity checks, through exploratory visualization, to business-facing recommendations grounded in evidence.